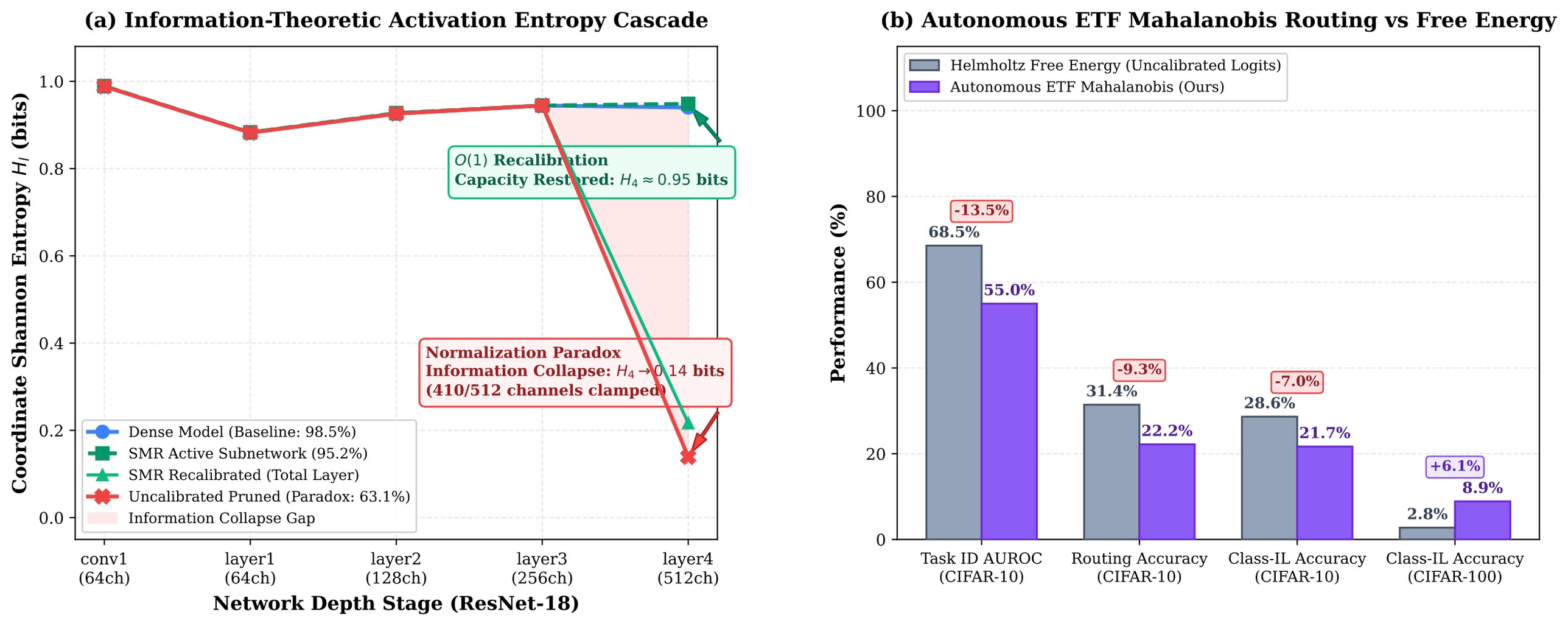
(a) Information-Theoretic Activation Entropy Cascade
Coordinate Shannon Entropy H_l (bits)
1.0
0.8
0.6
0.4
0.2
0.0
$O(1)$ Recalibration
Capacity Restored: $H_4 \approx 0.95$ bits
Normalization Paradox
Information Collapse: $H_4 \to 0.14$ bits
(410/512 channels clamped)
Dense Model (Baseline: 98.5%)
SMR Active Subnetwork (95.2%)
SMR Recalibrated (Total Layer)
Uncalibrated Pruned (Paradox: 63.1%)
Information Collapse Gap
conv1 (64ch)
layer1 (64ch)
layer2 (128ch)
layer3 (256ch)
layer4 (512ch)
Network Depth Stage (ResNet-18)
(b) Autonomous ETF Mahalanobis Routing vs Free Energy
Helmholtz Free Energy (Uncalibrated Logits)
Autonomous ETF Mahalanobis (Ours)
Performance (%)
100
80
60
40
20
0
-13.5%
68.5%
55.0%
-9.3%
31.4%
22.2%
-7.0%
28.6%
21.7%
+6.1%
2.8%
8.9%
Task ID AUROC (CIFAR-10)
Routing Accuracy (CIFAR-10)
Class-IL Accuracy (CIFAR-10)
Class-IL Accuracy (CIFAR-100)